

AYBERK YARKIN YILDIZ

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Summary

Researcher focusing on model-distributed inference via communication-aware pruning, mapping, and compression, and Markovian experimental design for robust re-training under concept drift. Experienced in building ML systems across sensing modalities, including wireless radar, RF signal sensing, sEMG motion analysis, and multivariate time-series modeling.

Education

Northeastern University <i>Ph.D. in Computer Engineering</i> <i>M.Sc. in Electrical and Computer Engineering</i>	<i>Boston, MA, USA</i> <i>Expected April 2028</i> <i>April 2025</i>
Bilkent University <i>B.Sc. in Electrical and Electronics Engineering</i>	<i>Ankara, Turkey</i> <i>July 2023</i>
Friedrich-Alexander-Universität <i>Erasmus Student in Elektrotechnik – Elektronik und Informationstechnik</i>	<i>Erlangen, Germany</i> <i>August 2022</i>
Mehmet Emin Resulzade <i>High School Degree</i>	<i>Ankara, Turkey</i> <i>June 2018</i>

Skills

Programming Languages: Python, MATLAB, C++
Frameworks: PyTorch, Tensorflow, PySpark, Slurm, Bash
Languages: English (Fluent – C1 level), Turkish (Native Fluency), German (Beginner – A2 level)
Selected Courses: Advanced Machine Learning, Verifiable Machine Learning, Probabilistic System Modeling, Reinforcement Learning and Decision Making Under Uncertainty, Advanced Computer Vision, Machine Learning with Small Data

Experience

Northeastern University <i>Research Assistant</i>	<i>Boston, MA, USA</i> <i>Sep 2023 – April 2028</i>
Research Labs: DNAL, WIoT, SPIRAL. Advisor: Prof. Stratis Ioannidis	
- Designed Markovian experimental design algorithms for optimal data selection to address performance degradation of ML models under concept drift, resulting in improved predictive accuracy with provable optimality guarantees. - Developed scalable, communication-aware optimization methods for neural network pruning and mapping to mitigate bandwidth, latency, and energy constraints in distributed systems, enabling resilient and low-latency inference across heterogeneous hardware topologies.	
Neurocess Limited <i>Data Science / Machine Learning Engineer (Remote)</i>	<i>London, UK</i> <i>Sep 2022 – July 2023</i>
Built Siamese CNN-Transformer architectures for multivariate sEMG time-series analysis to solve athlete performance monitoring under limited data, achieving similarity-based classification and anomaly detection in PyTorch and Keras.	
KOCLAB, National Magnetic Resonance Research Center (UMRAM) <i>Undergraduate Researcher</i>	<i>Ankara, Turkey</i> <i>Sep 2021 – Sep 2022</i>
Implemented deep learning models based on RNNs and Transformers to address missing data and temporal dependencies in biomedical time-series, producing imputation and forecasting pipelines validated in PyTorch experiments.	
TUBITAK SAGE <i>Intern</i>	<i>Ankara, Turkey</i> <i>June 2021 – July 2021</i>
Designed nano-drone hardware and software simulations in Altium Designer to improve agility and reduce detectability in defense applications, supporting early-stage validation of stealth-oriented aerial platforms.	
UMRAM <i>Intern</i>	<i>Ankara, Turkey</i> <i>July 2020 – Aug 2020</i>
Implemented and tested MATLAB-based interfaces for fundamental electronic devices (gaussmeter, analog filter, and DC power supply) to support scalable experimentation for future research, resulting in a publication in <i>Medical Physics</i> 🔗.	

Projects

Markovian Experimental Design under Concept Drift	<i>2024 – Present</i>
Implemented Kalman-filter-based data selection strategies to combat concept drift in sequential learning settings, demonstrating sustained predictive performance in ML models under non-stationary data distributions.	

- Communication-aware neural Mapping and Pruning Framework** 2024 – Present
- Developed a communication-aware pruning and model-mapping pipeline to reduce inference latency and network overhead in distributed CNNs, achieving up to $26\times$ speedup on real-world testbeds including Colosseum and Raspberry Pi clusters.
- Multi-Agent Distributed Inference over Large AI Models at the Network Edge** 2025
- Designed a compression-driven Model-Distributed Inference (MDI) framework to enable execution of large models across heterogeneous edge devices, reducing wireless communication overhead via lightweight pruning and quantization.
- Gradient Boosting Decision Trees on Medical Diagnosis** View publication [5]  2024
- Conducted large-scale empirical analysis of ensemble models to evaluate performance trade-offs against deep learning methods on tabular medical data, showing consistent superiority of GBDTs across multiple benchmarks.
- Wireless Radar Classification with Transformers** View preprint [4]  2023 – 2024
- Implemented Transformer-based radar signal classifiers with LoRA and conformal prediction to improve robustness under distribution shifts, achieving reliable OOD performance in PyTorch.
- Portable RF Signal Sensing System Using SDR** View publication [6]  2023
- Implemented a GPU-accelerated SDR-based Electronic Support Measures (ESM) system to address real-time RF signal awareness requirements, enabling efficient detection, measurement, and classification using GNU Radio and XGBoost.
- sEMG Motion Classification and Anomaly Detection** View preprint  2022 – 2023
- Developed few-shot Siamese learning models to address limited-label sEMG motion classification, achieving robust generalization across unseen motion patterns.
- Multivariate Time Series Imputation With Transformers** View publication [7]  2022
- Designed a Transformer-based autoencoder to solve missing-value imputation in multivariate time series, outperforming seven state-of-the-art methods by 13.5–50.5% over benchmark datasets.

Publications

[1] **A. Y. Yıldız**, L. Su, C. Joe-Wong, E. Yeh, and S. Ioannidis, “Markovian Experimental Design under Concept Drift.” (submitted to ICML 2026)

[2] **A. Y. Yıldız***, M. Sirera Perelló*, T. T. Davarakis*, J. Groen, H. Seferoglu, K. Chowdury, and S. Ioannidis, “CaMP: Communication-Aware Neural Mapping and Pruning.” (submitted to ACM SIGMETRICS 2026)

[3] **A. Y. Yıldız***, M. Sirera Perelló*, T. Li*, C. Li, P. Gholami, S. Reddy, T. T. Davarakis, A. Arora, K. Chowdhury, A. Eryilmaz, S. Ioannidis, H. Seferoglu, and N. Shroff, “Multi-Agent Distributed Inference over Large AI Models at the Network Edge.” (submitted to IEEE Wireless Communications Magazine)

[4] M. Belgiovine, J. Groen, M. Sirera, C. Tassie, **A. Y. Yıldız**, S. Ioannidis, and K. Chowdhury, “T-PRIME: Real-time Deployment of a Transformer-based Protocol Identification for Machine-learning at the Edge.” arXiv preprint arXiv:2401.04837 (2024). (submitted to IEEE/ACM Transactions on Networking)

[5] **A. Y. Yıldız** and A. Kalayci, “Gradient boosting decision trees on medical diagnosis over tabular data.” 2025 IEEE International Conference on AI and Data Analytics (ICAD). IEEE, 2025.

[6] G. S. Yavuz, B. Saygılı, Y. Aydınli, R. Dalkıran, İ. Eşin, M. Uluçay, B. Uykulu, S. S. Kıyma, O. Arikan, and **A. Y. Yıldız**, “Detection and classification architecture for sdr based radar electronic support measure systems.” 2024 32nd Signal Processing and Communications Applications Conference (SIU). IEEE, 2024.

[7] **A. Y. Yıldız**, E. Koç, and A. Koç. “Multivariate time series imputation with transformers.” IEEE Signal Processing Letters 29 (2022): 2517-2521.

Honors & Awards

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- Invited Talks: IEEE SPS, IEEE ICIP 2023 – 2024
 - IEEE SPS’s top 25 downloaded articles 2022 – 2023
 - Research Excellence Award at Bilkent University 2023
 - High Honor Student & Tuition Scholarship at Bilkent University 2018 – 2023
 - 8/8th grade practical and theoretical certificate for piano, ABRSM, London, UK 2023

Services

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- Technical Reviewer: AISTATS 2026, IEEE ICAD 2026
 - Amazon IET-MSI Program Participant Oct 2024
 - Northeastern University SPIRAL Committee Member Dec 2024
 - Organized seminars with participants from universities in USA and Switzerland.